KCET-2017 (Mathematics)

1. If A and B are finite sets and $A \subset B$, then-
   a. $n(A \cup B) = n(A)$
   b. $n(A \cap B) = n(B)$
   c. $n(A \cup B) = n(B)$
   d. $n(A \cap B) = \phi$

2. The value of $\cos^2 45^0 - \sin^2 15^0$ is -
   a. $\frac{\sqrt{3}}{2}$
   b. $\frac{\sqrt{3}}{4}$
   c. $\frac{\sqrt{3}+1}{2\sqrt{2}}$
   d. $\frac{\sqrt{3}-1}{2\sqrt{2}}$

3. 3 + 5 + 7 + ... to n term is -
   a. n(n + 2)
   b. n(n - 2)
   c. n^2
   d. $(n + 1)^2$

4. If $\left(\frac{1+i}{1-i}\right)^m = 1$, then the least positive integral value of m is -
   a. 2
   b. 3
   c. 4
   d. 1

5. If $|x-2| \leq 1$, then -
   a. $x \in [1, 3]$
   b. $x \in (1, 3)$
   c. $x \in [-1, 3)$
   d. $x \in [-1, 3)$

6. If ${}^nC_{12} = {}^nC_8$ then n is equal to -
   a. 26
   b. 12
   c. 6
   d. 20

7. The total number of terms in the expansion of $(x + a)^{47} - (x - a)^{47}$ after simplification is -
   a. 24
   b. 47
   c. 48
   d. 96

8. Equation of line passing through the point (1, 2) and perpendicular to the line $y = 3x - 1$ is :-
   a. $x + 3y - 7 = 0$
   b. $x + 3y + 7 = 0$
   c. $x + 3y = 0$
   d. $x - 3y = 0$

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9. The eccentricity of the ellipse $\frac{x^2}{36}+\frac{y^2}{16}=1$ is -

   a. $\frac{2\sqrt{5}}{6}$ b. $\frac{2\sqrt{5}}{4}$

   c. $\frac{2\sqrt{13}}{6}$ d. $\frac{2\sqrt{13}}{4}$

10. The perpendicular distance of the point P(6, 7, 8) from XY-plane is -

    a. 8 b. 7

    c. 6 d. 5

11. The value of $\lim_{\theta\to 0}\frac{1-\cos 4\theta}{1-\cos 6\theta}$ is -

    a. 4/9 b. 9/4

    c. 9/3 d. 3/4

12. The contrapositive statement of the statement "If x is prime number, then x is odd" is -

    a. If x is not a prime number, then x is not odd

    b. If x is a prime number, then x is not odd

    c. If x is not a prime number, then x is odd

    d. If x is not odd, then x is not a prime number

13. If coefficient of variation is 60 and standard deviation is 24, then Arithmetic mean is -

    a. 40 b. 7/20

    c. 20/7 d. 1/40

14. The range of the function $f(x)=\sqrt{9-x^2}$ is -

    a. (0, 3) b. [0, 3]

    c. (0, 3] d. [0, 3)

15. Let f : R $\to$ R be defined by $f(x)=x^4$, then -

    a. f is one-one and onto b. f may be one-one and onto

    c. f is one-one but not onto d. f is neither one-one nor onto

16. The range of $\sec^{-1}x$ is -

    a. $\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$ b. $\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$

    c. $[0,\pi]$ d. $[0,\pi]-\left\{\frac{\pi}{2}\right\}$

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17. If $\tan^{-1}x+\tan^{-1}y=\frac{4\pi}{5}$, then $\cot^{-1}x+\cot^{-1}y$ is equal to -

a. π b. $\frac{\pi}{5}$

c. $\frac{2\pi}{5}$ d. $\frac{3\pi}{5}$

18. If $f(x)=8x^3, g(x)=x^{1/3}$, Then fog (x) is -

a. $8x$ b. 8^3x

c. $(8x)^{1/3}$ d. $8x^3$

19. If $A=\frac{1}{\pi}\begin{bmatrix}\sin^{-1}(\pi x) & \tan^{-1}\left(\frac{x}{\pi}\right)\\ \sin^{-1}\left(\frac{x}{\pi}\right) & \cot^{-1}(\pi x)\end{bmatrix}$, $B=\begin{bmatrix}-\cos^{-1}(\pi x) & \tan^{-1}\left(\frac{x}{\pi}\right)\\ \sin^{-1}\left(\frac{x}{\pi}\right) & -\tan^{-1}(\pi x)\end{bmatrix}$ then A – B is equal to -

a. I b. 0

c. 2I d. $\frac{1}{2}I$

20. If a matrix A is both symmetric and skew symmetric, then -

a. A is diagonal matrix b. A is zero matrix

c. A is scalar matrix d. A is square matrix

21. If $2\begin{bmatrix}1 & 3\\ 0 & x\end{bmatrix}+\begin{bmatrix}y & 0\\ 1 & 2\end{bmatrix}=\begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix}$, then the value of x and y are -

a. $x=3,\ y=3$ b. $x=-3,\ y=3$

c. $x=3,\ y=-3$ d. $x=-3,\ y=-3$

22. Binary operation $*$ on R – {-1} defined by a $*$ b = $\frac{a}{b+1}$ is -

a. $*$ is associative and commutative

b. $*$ is associative but not commutative

c. $*$ is neither associative nor commutative

d. $*$ is commutative but not associative

23. If $\begin{vmatrix}3 & x\\ x & 1\end{vmatrix}=\begin{vmatrix}3 & 2\\ 4 & 1\end{vmatrix}$ then x is equal to -

a. 2 b. 4

c. 8 d. $\pm 2\sqrt{2}$

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KCET-2017 (Mathematics)

24. If A is a square matrix of order 3×3, then $|KA|$ is equal to -

a. $K|A|$ b. $K^2|A|$

c. $K^3|A|$ d. $3K|A|$

25. The area of triangle with vertices (K, 0), (4, 0), (0, 2) is 4 square units, then value of K is -

a. 0 or 8 b. 0 or -8

c. 0 d. 8

26. Let $\Delta=\begin{vmatrix} Ax & x^2 & 1 \\ By & y^2 & 1 \\ Cz & z^2 & 1 \end{vmatrix}$ and $\Delta_1=\begin{vmatrix} A & B & C \\ x & y & z \\ zy & zx & xy \end{vmatrix}$ then

a. $\Delta_1=-\Delta$ b. $\Delta_1=\Delta$

c. $\Delta_1\neq\Delta$ d. $\Delta_1=2\Delta$

27. If $f(x)=\begin{cases} Kx^2 & \text{if } x\le 2 \\ 3 & \text{if } x>2 \end{cases}$ is continuous at $x=2$, then the value of K is -

a. 3 b. 4

c. 3/4 d. 4/3

28. The value of C in Mean value theorem for the function $f(x)=x^2$ in [2, 4] is -

a. 3 b. 2

c. 4 d. 7/2

29. The point on the curve $y^2=x$ where the tangent makes an angle of $\pi/4$ with X-axis is -

a. $\left(\frac{1}{2},\frac{1}{4}\right)$ b. $\left(\frac{1}{4},\frac{1}{2}\right)$

c. $(4,\ 2)$ d. $(1,\ 1)$

30. The function $f(x)=x^2+2x-5$ is strictly increasing in the interval -

a. $(-1,\ \infty)$ b. $(-\infty,-1)$

c. $[-1,\ \infty)$ d. $(-1,\ \infty]$

31. The rate of change of a sphere with respect to its surface area when the radius is 4 cm is -

a. $4\ cm^3/cm^2$ b. $2\ cm^3/cm^2$

c. $6\ cm^3/cm^2$ d. $8\ cm^3/cm^2$

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32. If $y=\tan^{-1}\left(\frac{\sin x+\cos x}{\cos x-\sin x}\right)$, then $\frac{dy}{dx}$ is equal to -

a. 1/2
b. $\pi/4$
c. 0
d. 1

33. If $y=\begin{vmatrix} f(x) & g(x) & h(x) \\ 1 & m & n \\ a & b & c \end{vmatrix}$, then $\frac{dy}{dx}$, is equal to -

a. $\begin{vmatrix} f'(x) & g'(x) & h'(x) \\ 1 & m & n \\ a & b & c \end{vmatrix}$
b. $\begin{vmatrix} 1 & m & n \\ f'(x) & g'(x) & h'(x) \\ a & b & c \end{vmatrix}$
c. $\begin{vmatrix} f'(x) & 1 & a \\ g'(x) & m & b \\ h'(x) & n & c \end{vmatrix}$
d. $\begin{vmatrix} 1 & m & n \\ a & b & c \\ f'(x) & g'(x) & h'(x) \end{vmatrix}$

34. If $\sin x=\frac{2t}{1+t^2}, \tan y=\frac{2t}{1-t^2},$ then $\frac{dy}{dx}$ is equal to -

a. 1
b. 0
c. -1
d. 2

35. The derivative of $\cos^{-1}\left(2x^2-1\right)$ w.r.t $\cos^{-1}x$ is -

a. 2
b. $\frac{-1}{2\sqrt{1-x^2}}$
c. $\frac{2}{x}$
d. $1-x^2$

36. If $y=\log(\log x)$ then $\frac{d^2y}{dx^2}$ is equal to -

a. $\frac{-(1+\log x)}{(x\log x)^2}$
b. $\frac{-(1+\log x)}{x^2\log x}$
c. $\frac{(1+\log x)}{(x\log x)^2}$
d. $\frac{(1+\log x)}{x^2\log x}$

37. $\int\frac{(x+3)e^x}{(x+4)^2}dx$ is equal to -

a. $\frac{1}{(x+4)^2}+C$
b. $\frac{e^x}{(x+4)^2}+C$
c. $\frac{e^x}{(x+4)}+C$
d. $\frac{e^x}{(x+3)}+C$

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38. $\int \frac{\cos 2x - \cos 2\theta}{\cos x - \cos \theta}\, dx$ is equal to –

a. $2(\sin x + x\cos\theta) + C$
b. $2(\sin x - x\cos\theta) + C$
c. $2(\sin x + 2x\cos\theta) + C$
d. $2(\sin x - 2x\cos\theta) + C$

39. $\int \sqrt{x^2 + 2x + 5}\, dx$ is equal to -

a. $\frac{1}{2}(x+1)\sqrt{x^2+2x+5} + 2\log\left|x+1+\sqrt{x^2+2x+5}\right| + C$
b. $(x+1)\sqrt{x^2+2x+5} + 2\log\left|x+1+\sqrt{x^2+2x+5}\right| + C$
c. $(x+1)\sqrt{x^2+2x+5} - 2\log\left|x+1+\sqrt{x^2+2x+5}\right| + C$
d. $(x+1)\sqrt{x^2+2x+5} + \frac{1}{2}\log\left|x+1+\sqrt{x^2+2x+5}\right| + C$

40. $\int_0^{\pi/2} \frac{\tan^7 x}{\cot^7 x + \tan^7 x}\, dx$ is equal to -

a. $\frac{\pi}{2}$
b. $\frac{\pi}{4}$
c. $\frac{\pi}{6}$
d. $\frac{\pi}{3}$

41. $\int_{-5}^{5} |x+2|\, dx$ is equal to -

a. 29
b. 28
c. 27
d. 30

42. $\int_{-\pi/2}^{\pi/2} \frac{dx}{e^{\sin x} + 1}$ is equal to -

a. 0
b. 1
c. $-\frac{\pi}{2}$
d. $\frac{\pi}{2}$

43. $\int_0^{\pi/2} \frac{1}{a^2 \cdot \sin^2 x + b^2 \cdot \cos^2 x}\, dx$ is equal to -

a. $\frac{\pi a}{4b}$
b. $\frac{\pi a}{2b}$
c. $\frac{\pi b}{4a}$
d. $\frac{\pi}{2ab}$

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44. The area of the region bounded by the curve $y = x^2$ and the line $y = 16$ is -

a. $\frac{32}{3}$ sq. units
b. $\frac{256}{3}$ sq. units
c. $\frac{64}{3}$ sq. units
d. $\frac{128}{3}$ sq. units

45. Area of the region bounded by the curve $y = \cos x$, $x = 0$ and $x = \pi$ is -

a. 2 sq. units
b. 4 sq. units
c. 3 sq. units
d. 1 sq. unit

46. The degree of the differential equation $\left[1+\left(\frac{dy}{dx}\right)^2\right]^2 = \frac{d^2y}{dx^2}$ is -

a. 1
b. 2
c. 3
d. 4

47. General solution of differential equation $\frac{dy}{dx} + y = 1 (y \neq 1)$ is -

a. $\log\left|\frac{1}{1-y}\right| = x + C$
b. $\log|1-y| = x + C$
c. $\log|1+y| = x + C$
d. $\log\left|\frac{1}{1-y}\right| = -x + C$

48. The integrating factor of the differential equation $x.\frac{dy}{dx} + 2y = x^2$ is $(x \neq 0)$ -

a. x^2
b. $\log|x|$
c. $e^{\log x}$
d. x

49. If $\vec{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$ are orthogonal, then value of λ is -

a. 0
b. 1
c. $\frac{3}{2}$
d. $-\frac{5}{2}$

50. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, then the value of $\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}$ is equal to -

a. 1
b. 3
c. $-\frac{3}{2}$
d. $\frac{3}{2}$

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KCET-2017 (Mathematics)

51. If $\vec{a}$ & $\vec{b}$ are unit vectors, the angle between $\vec{a}$ and $\vec{b}$ for $\sqrt{3}\,\vec{a}-\vec{b}$ to be unit vector is -
    a. 30°
    b. 45°
    c. 60°
    d. 90°

52. Reflexion of the point (α,β,γ) in XY plane is -
    a. $(\alpha,\beta,0)$
    b. $(0,0,\gamma)$
    c. $(-\alpha,-\beta,\gamma)$
    d. $(\alpha,\beta,-\gamma)$

53. The plane $2x-3y+6z-11=0$ makes an angle $\sin^{-1}(\alpha)$ with X-axis. The value of α is equal to -
    a. $\frac{\sqrt{3}}{2}$
    c. $\frac{\sqrt{2}}{3}$
    b. $\frac{2}{7}$
    d. $\frac{3}{7}$

54. The distance of the point (-2, 4, -5) from the line $\frac{x+3}{3}=\frac{y-4}{5}=\frac{z+8}{6}$ is -
    a. $\frac{\sqrt{37}}{10}$
    b. $\sqrt{\frac{37}{10}}$
    c. $\frac{37}{\sqrt{10}}$
    d. $\frac{37}{10}$

55. A box has 100 pens of which 10 are defective. The probability that out of a sample of 5 pens drawn one by one with replacement and at most one is defective is -
    a. $\frac{9}{10}$
    b. $\frac{1}{2}\left(\frac{9}{10}\right)^4$
    c. $\left(\frac{9}{10}\right)^5+\frac{1}{2}\left(\frac{9}{10}\right)^4$
    d. $\frac{1}{2}\left(\frac{9}{10}\right)^5$

56. Two events A and B will be independent if -
    a. A and B are mutually exclusive
    b. $P(A'\cap B')=(1-P(A))(1-P(B))$
    c. $P(A)=P(B)$
    d. $P(A)+P(B)=1$

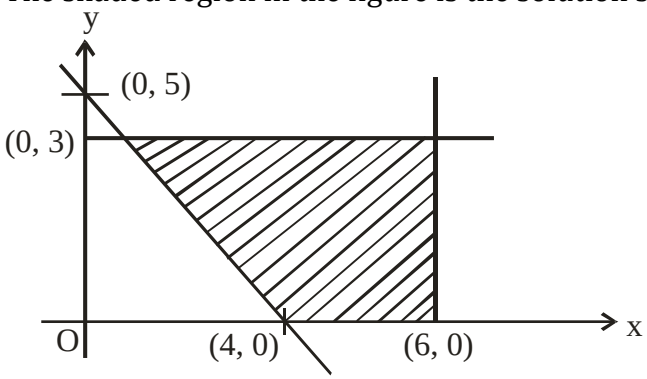
57. The probability distribution of X is

X	0	1	2	3
P(X)	0.3	k	2k	2k

The value of k is -
    a. 0.14
    b. 0.3
    c. 0.7
    d. 1

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58. The shaded region in the figure is the solution set of the in equations –



a. $5x+4y \geq 20, x \leq 6, y \geq 3, x \geq 0, y \geq 0$

b. $5x+4y \leq 20, x \leq 6, y \leq 3, x \geq 0, y \geq 0$

c. $5x+4y \geq 20, x \leq 6, y \leq 3, x \geq 0, y \geq 0$

d. $5x+4y \geq 20, x \geq 6, y \leq 3, x \geq 0, y \geq 0$

59. If an LPP admits optimal solution at two consecutive vertices of a feasible region, then -

a. The required optimal solution is at the midpoint of the line joining two points

b. The optimal solution occurs at every point on the line joining these two points

c. The LPP under consideration is not solvable

d. The LPP under consideration must be reconstructed

60. $\int_{0.2}^{3.5}[x]\,dx$ equal to –

a. 4

b. 4.5

c. 3.5

d. 3

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KCET-2017 (Mathematics)

ANSWER KEYS

1. (c)	2. (b)	3. (a)	4. (c)	5. (a)	6. (d)	7. (a)	8. (a)	9. (a)	10. (a)
11. (a)	12. (d)	13. (a)	14. (b)	15. (d)	16. (d)	17. (b)	18. (a)	19. (G)	20. (b)
21. (a)	22. (c)	23. (d)	24. (c)	25. (a)	26. (b)	27. (c)	28. (a)	29. (b)	30. (a)
31. (b)	32. (d)	33. (a,c,d)	34. (a)	35. (a)	36. (a)	37. (c)	38. (a)	39. (a)	40. (b)
41. (a)	42. (d)	43. (d)	44. (b)	45. (a)	46. (a)	47. (a)	48. (a)	49. (d)	50. (c)
51. (a)	52. (d)	53. (c)	54. (b)	55. (c)	56. (b)	57. (a)	58. (c)	59. (b)	60. (b)

* G – Indicates One GRACE MARK awarded for the question number

BYJU'S

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KCET-2017 (Mathematics)

Solution

1. (c)

Consider the function $A \subset B$

$A \cap B = A$

$N(A \cap B) = n(A)$

Also,

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= n(A) + n(B) - n(A)$

$= n(B)$

2. (b)

$\cos^2 45^0 - \sin^2 15^0$

$\Rightarrow \cos(45^0 + 15^0)\cos(45^0 - 15^0) \qquad \{\because \cos^2 A - \sin^2 B = \cos(A+B)\cos(A-B)\}$

$= \cos 60^0 \ \cos 30^0$

$= \left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right)$

$= \frac{\sqrt{3}}{4}$

3. (a)

3+5+7+……+n

(a=3, d=2)

$S_n = \frac{n}{2}[2a + (n-1)d]$

$S_n = \frac{n}{2}[2(3) + (n-1)(2)]$

$S_n = \frac{n}{2}[6 + 2n - 2]$

$S_n = \frac{n}{2}[4 + 2n]$

$S_n = n(n+2)$

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4. (c)

Consider the equation

$\left(\frac{1+i}{1-i}\right)^m = 1$

$\Rightarrow \left(\frac{1+i}{1-i}\times\frac{1+i}{1+i}\right)^m = 1$

$\Rightarrow \left(\frac{(1+i)^2}{1-i^2}\right)^m = 1 \qquad \{\because i^2 = -1\}$

$\Rightarrow \left(\frac{1+2i-1}{1+1}\right)^m = 1$

$\Rightarrow \left(\frac{2i}{2}\right)^m = 1$

$\Rightarrow (i)^m = 1 \qquad \{\because i^4 = 1\}$

$\Rightarrow (i)^m = 1^4$

$\Rightarrow m = 4$

5. (a)

$|x-2| \le 1$

$-1 \le x - 2 \le 1$

$-1 + 2 \le x \le 1 + 2$

$-1 \le x \le 3$

Thus, $x \in [1, 3]$

6. (d)

It is known that if,

${}^nC_x = {}^nC_y$

Then $x + y = n$

Now ${}^nC_{12} = {}^nC_8$

So $12 + 8 = n$

$n = 20$

7. (a)

Consider the expansion $(x + a)^{47} - (x - a)^{47}$

Here, n = 47 which is odd

The total number of term in the expansion is,

Total term $= \frac{n+1}{2}$

$= \frac{47+1}{2} = \frac{48}{2} = 24$

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8. (a)

The equation of line $y = 3x - 1$

Slope of the line is $m = 3$

The slope of the line perpendicular to the given line $m_1 = -\frac{1}{m}$

$$m_1 = -\frac{1}{3}$$

The equation of required line is

$(y - y_1) = m_1(x - x_1)$

$\Rightarrow (y - 2) = -\frac{1}{3}(x - 1)$

$\Rightarrow 3y - 6 = -x + 1$

$x + 3y - 7 = 0$

9. (a)

The equation of ellipse

$\Rightarrow \frac{x^2}{36} + \frac{y^2}{16} = 1$

$\Rightarrow \frac{x^2}{(6)^2} + \frac{y^2}{(4)^2} = 1$

So, $a = 6$

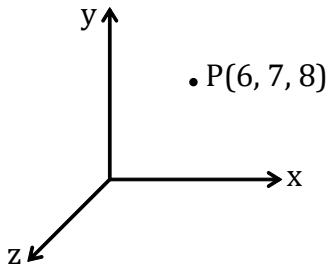
$b = 4$

Eccentricity $(e) = \sqrt{1 - \frac{b^2}{a^2}}$

$$= \sqrt{1 - \frac{4^2}{6^2}} = \sqrt{1 - \frac{16}{36}} = \sqrt{\frac{20}{36}} = \frac{2\sqrt{5}}{6}$$

10. (a)

The given point is P (6, 7, 8)



Distance of any point from xy-plane = |z|

D = |z|

D = |8|

D = 8

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11. (a)

Let $L = \lim_{\theta \to 0} \frac{1-\cos 4\theta}{1-\cos 6\theta}$

$= \lim_{\theta \to 0} \frac{1-1+2\sin^2 2\theta}{1-1+2\sin^2 3\theta}$ $\{\because \cos 2A = 1 - 2\sin^2 A\}$

$= \lim_{\theta \to 0} \frac{2\sin^2 2\theta}{2\sin^2 3\theta}$

$= \lim_{\theta \to 0} \left(\frac{\sin 2\theta}{2\theta}\right)^2 \left(\frac{3\theta}{\sin 3\theta}\right)^2 \left(\frac{4}{9}\right)$

$= (1)\cdot(1)\cdot\left(\frac{4}{9}\right) = \frac{4}{9}$

12. (d)

The contrapositive statement of the statement "If x is a prime number then x is odd" is "If x is not odd then x is not a prime number.

13. (a)

The coefficient of variation is CV = 60 and the standard deviation is $\sigma = 24$

Coefficient of variation $= \frac{\sigma}{\bar{x}} \times 100$

$60 = \frac{24}{\bar{x}} \times 100$

$\bar{x} = \frac{24}{60} \times 100$

$\bar{x} = 40$

14. (b)

Let $y = f(x) = \sqrt{9-x^2}$

Squaring both side

$y^2 = 9 - x^2$

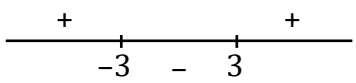
$x = \sqrt{9-y^2}$

Now,

$9 - y^2 \geq 0$

$\Rightarrow (3 - y)(3 + y) \geq 0$

$\Rightarrow (y - 3)(y + 3) \leq 0$



$y \in [0, 3]$

So $f(x) \in [0, 3]$

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15. (d)

Given function $f(x) = x^4$

Let $f(x_1) = x_1^4$

And

$F(x_2) = x_2^4$

So, $f(x_1) = f(x_2)$

$x_1^4 = x_2^4$

$x_1 = \pm x_2$

So f(x) is not one-one.

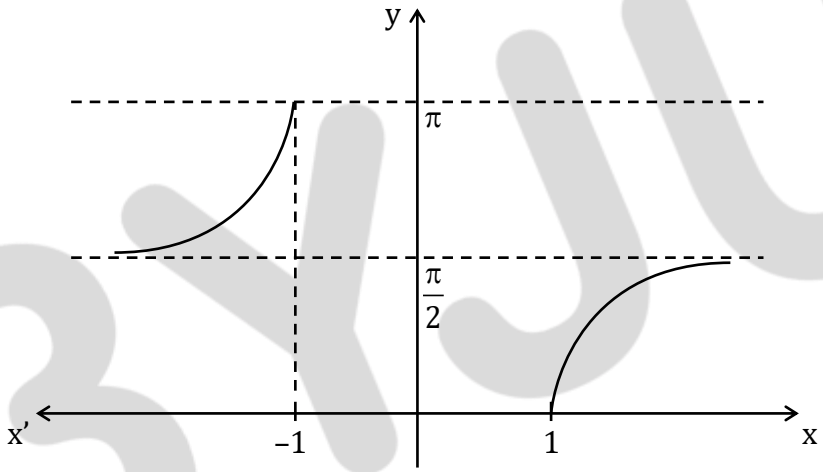
The co-domain of f(x) is R and the range is $f(x) \in [0, \infty)$

Since co-domain is not equal to the range of the function.

Thus, the function is not onto.

16. (d)

The graph of $\sec^{-1} x$



$y = \sec^{-1} x$

The range of the $\sec^{-1} x$ is $[0, \pi] - \left\{\frac{\pi}{2}\right\}$

17. (b)

Given function

$$\tan^{-1} x + \tan^{-1} y = \frac{4\pi}{5}$$

$$\Rightarrow \quad \frac{\pi}{2} - \cot^{-1} x + \frac{\pi}{2} - \cot^{-1} y = \frac{4\pi}{5} \qquad \left\{Q \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}\right\}$$

$$\Rightarrow \quad \cot^{-1} x + \cot^{-1} y = \pi - \frac{4\pi}{5}$$

$$\cot^{-1} x + \cot^{-1} y = \frac{\pi}{5}$$

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18. (a)

Given function

$f(x) = 8x^3$

And $g(x) = x^{1/3}$

Now, $fog(x) = f[g(x)]$

$= f[x^{1/3}]$

$= 8(x^{1/3})^3$

$= 8x$

19. (G)

Bonus

20. (b)

Let A be a matrix that is both symmetric and skew symmetric

Therefore,

$A' = A$

And $A' = -A$

This case is only possible when A is a zero matrix.

21. (a)

$$2\begin{bmatrix}1 & 3\\ 0 & x\end{bmatrix} + \begin{bmatrix}y & 0\\ 1 & 2\end{bmatrix} = \begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix}$$

$$\begin{bmatrix}2 & 6\\ 0 & 2x\end{bmatrix} + \begin{bmatrix}y & 0\\ 1 & 2\end{bmatrix} = \begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix}$$

$$\begin{bmatrix}2+y & 6\\ 1 & 2x+2\end{bmatrix} = \begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix}$$

So

$2 + y = 5$ | $2x + 2 = 8$

$y = 3$ | $2x = 6$

$x = 3$

22. (c)

Given that

$$a * b = \frac{a}{b+1} \qquad \ldots(1)$$

So

$$b * a = \frac{b}{a+1} \qquad \ldots(2)$$

Thus

$a * b \neq b * a$

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Hence, the operation is not commutative.

Now,

$$a * (b * c) = a * \left(\frac{b}{c+1}\right)$$

$$= \frac{a}{\left(\frac{b}{c+1}\right)+1}$$

$$= \frac{a(c+1)}{b+c+1}$$

and

$$(a * b) * c = \left(\frac{a}{b+1}\right) * c$$

$$= \frac{\frac{a}{b+1}}{c+1}$$

$$= \frac{a}{(b+1)(c+1)}$$

So

$$a * (b * c) = (a * b) * c$$

Hence, the operation is not associative.

23. (d)

$$\begin{vmatrix} 3 & x \\ x & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix}$$

$\Rightarrow 3 - x^2 = 3 - 8$

$\Rightarrow -x^2 = -8$

$\Rightarrow x^2 = 8$

$\Rightarrow x = \pm 2\sqrt{2}$

24. (c)

For a n × n order matrix of degree 2.

$|KA| = K^n |A|$

Here n is the order of the matrix.

Since the matrix of the 3 × 3

$|KA| = K^3 |A|$

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25. (a)

The vertices of the triangle are (K, 0), (4, 0), (0, 2)

The area of the triangle is 4 sq. units.

$$\Rightarrow \quad \Delta = \frac{1}{2}\begin{vmatrix} K & 0 & 1 \\ 4 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix}$$

$$\Rightarrow \quad 4 = \frac{1}{2}|\,[K(0-2) - 0(4-0) + 1(8-0)]|$$

$$\Rightarrow \quad 4 = \frac{1}{2}|\,[-2K + 8]| \Rightarrow \; 8 = |-2K + 8|$$

$$-2K + 8 = \pm 8$$

Now,

$$-2K + 8 = 8$$

$$-2K = 0$$

$$K = 0$$

And

$$-2K + 8 = -8$$

$$-2K = -16; \quad K = 8$$

26. (b)

$$\Delta = \begin{vmatrix} Ax & x^2 & 1 \\ By & y^2 & 1 \\ Cz & z^2 & 1 \end{vmatrix}$$

And $$\Delta_1 = \begin{vmatrix} A & B & C \\ x & y & z \\ zy & zx & xy \end{vmatrix}$$

Apply $C_1 \to xC_1$, $C_2 \to yC_2$ and $C_3 \to zC_3$

$$\Delta_1 = \frac{1}{xyz}\begin{vmatrix} Ax & By & Cz \\ x^2 & y^2 & z^2 \\ xyz & xyz & xyz \end{vmatrix}$$

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$$\Delta_1 = \frac{xyz}{xyz}\begin{vmatrix} Ax & By & Cz \\ x^2 & y^2 & z^2 \\ 1 & 1 & 1 \end{vmatrix}$$

$$\Delta_1 = \begin{vmatrix} Ax & By & Cz \\ x^2 & y^2 & z^2 \\ 1 & 1 & 1 \end{vmatrix}$$

Since

$|A'| = |A|$

So,

$$\Delta_1 = \begin{vmatrix} Ax & x^2 & 1 \\ By & y^2 & 1 \\ Cz & z^2 & 1 \end{vmatrix}$$

$\Delta_1 = \Delta$

27. (c)

$$f(x) = \begin{cases} kx^2 & ; x \le 2 \\ 3 & ; x > 2 \end{cases}$$

The function is continuous at x = 2. So,

$\lim_{x\to 2} Kx^2 = \lim_{x\to 2} 3$

$K(2)^2 = 3$

$K = \frac{3}{4}$

28. (a)

The mean value theorem is given as

$f'(c) = \frac{f(b)-f(a)}{b-a}$

Now,

$f(x) = x^2$

$f'(x) = 2x$

Therefore

$f'(c) = \frac{f(4)-f(2)}{4-2}$

$\Rightarrow \quad 2c = \frac{16-4}{2}$

$\Rightarrow \quad c = \frac{12}{4}$

$c = 3$

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29. (b)

The curve $y^2 = x$

Differentiate with respect to x

$$2y\frac{dy}{dx} = 1$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{1}{2y}$$

$$\Rightarrow \quad \tan\theta = \frac{1}{2y} \qquad \left\{Q\,\theta = \frac{\pi}{4}\right\}$$

$$\Rightarrow \quad \tan\frac{\pi}{4} = \frac{1}{2y}$$

$$\Rightarrow \quad 1 = 2y$$

$$\Rightarrow \quad y = \frac{1}{2}$$

Thus,

$$\left(\frac{1}{2}\right)^2 = x \qquad \left\{Q\,y^2 = x \text{ and } y = \frac{1}{2}\right\}$$

$$x = \frac{1}{4}$$

The required point is $\left(\frac{1}{4}, \frac{1}{2}\right)$

30. (a)

$f(x) = x^2 + 2x - 5$

Differentiate with respect to x

$f'(x) = 2x + 2$

Now, function is strictly increasing

$f'(x) > 0$

$2(x + 1) > 0$

$x + 1 > 0$

$x > -1$

The function is increasing in $(-1, \infty)$

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31. (b)

The volume of the sphere is

$v = \frac{4}{3}\pi a^3$

Differentiate with respect to a

$\frac{dv}{da} = \frac{4}{3}\pi (3a^2)$

$\frac{dv}{da} = 4\pi a^2$ ... (1)

The surface area of the sphere is

$s = 4\pi a^2$

Differentiate with respect to a

$\frac{ds}{da} = 4\pi(2a)$

$\frac{ds}{da} = 8\pi a$ ... (2)

Now, divide eq(1) by eq(2)

$\frac{dv}{ds} = \frac{dv}{da} \times \frac{da}{ds}$

$\Rightarrow \frac{dv}{ds} = \frac{4\pi a^2}{8\pi a}$

$\Rightarrow \frac{dv}{ds} = \frac{a}{2}$

$\Rightarrow \frac{dv}{ds} = \frac{4}{2}$ {$\because$ radius (a) = 4}

$\Rightarrow \frac{dv}{ds} = 2\ cm^3/cm^2$

32. (d)

$y = \tan^{-1}\left(\frac{\sin x + \cos x}{\cos x - \sin x}\right)$

$y = \tan^{-1}\left(\frac{1 + \tan x}{1 - \tan x}\right)$

$y = \tan^{-1}\left[\tan\left(\frac{\pi}{4} + x\right)\right]$

$y = \frac{\pi}{4} + x$

Differentiate with respect to x

$\frac{dy}{dx} = 1$

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33. (a,c,d)

$$y = \begin{vmatrix} f(x) & g(x) & h(x) \\ l & m & n \\ a & b & c \end{vmatrix}$$

Differentiate with respect to x

$$\frac{dy}{dx} = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ l & m & n \\ a & b & c \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ \frac{d(l)}{dx} & \frac{d(m)}{dx} & \frac{d(n)}{dx} \\ a & b & c \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ l & m & n \\ \frac{d(a)}{dx} & \frac{d(b)}{dx} & \frac{d(c)}{dx} \end{vmatrix}$$

$$\frac{dy}{dx} = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ l & m & n \\ a & b & c \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ 0 & 0 & 0 \\ a & b & c \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ l & m & n \\ 0 & 0 & 0 \end{vmatrix}$$

$$\frac{dy}{dx} = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ l & m & n \\ a & b & c \end{vmatrix} = \begin{vmatrix} l & m & n \\ a & b & c \\ f'(x) & g'(x) & h'(x) \end{vmatrix}$$

$$\frac{dy}{dx} = \begin{vmatrix} f'(x) & l & a \\ g'(x) & m & b \\ h'(x) & n & c \end{vmatrix}$$

34. (a)

$$\sin x = \frac{2t}{1+t^2}$$

$$x = \sin^{-1}\left(\frac{2t}{1+t^2}\right)$$

$$x = 2\tan^{-1} t \qquad \ldots (1)$$

And

$$\tan y = \frac{2t}{1-t^2}$$

$$y = \tan^{-1}\left(\frac{2t}{1-t^2}\right)$$

$$y = 2\tan^{-1} t \qquad \ldots (2)$$

From equation (1) and (2)

$$y = x$$

$$\frac{dy}{dx} = 1$$

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35. (a)

Let $v = \cos^{-1}(2x^2 - 1)$ and $u = \cos^{-1} x$

$v = \cos^{-1}(2x^2 - 1)$

$v = 2\cos^{-1} x$

Differentiate with respect to x

$$\frac{dv}{dx} = \frac{-2}{\sqrt{1-x^2}}$$

And

$u = \cos^{-1} x$

$$\frac{du}{dx} = \frac{-1}{\sqrt{1-x^2}}$$

Now,

$$\frac{dv}{du} = \frac{du}{dx} \times \frac{dx}{dv}$$

$$= \frac{(-2)}{\sqrt{1-x^2}} \times \frac{\sqrt{1-x^2}}{(-1)}$$

$= 2$

36. (a)

$y = \log(\log x)$

Differentiate with respect to x

$$\frac{dy}{dx} = \frac{1}{x} \times \left(\frac{1}{\log x}\right)$$

$$\frac{dy}{dx} = (x \log x)^{-1}$$

Differentiate with respect to x

$$\frac{d^2y}{dx^2} = (-1)(x \log x)^{-2}\left[\log x + x\left(\frac{1}{x}\right)\right]$$

$= -(x \log x)^{-2}(\log x + 1)$

$$= \frac{-(\log x + 1)}{(x \log x)^2}$$

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37. (c)

Let $I = \int \frac{(x+3)e^x}{(x+4)^2} dx$

$I = \int e^x \left[\frac{x+4-1}{(x+4)^2}\right] dx$

$I = \int e^x \left[\frac{1}{x+4} - \frac{1}{(x+4)^2}\right] dx$

$I = \int e^x \left(\frac{1}{x+4}\right) dx - \int e^x \left(\frac{1}{x+4}\right)^2 dx$

$I = \frac{e^x}{x+4} - \int e^x \left[\frac{-1}{(x+4)^2}\right] dx - \int e^x \left[\frac{1}{(x+4)^2}\right] dx$

$I = \frac{e^x}{x+4} + \int \frac{e^x}{(x+4)^2} dx - \int \frac{e^x}{(x+4)^2} dx$

$I = \frac{e^x}{x+4} + C$

38. (a)

$I = \int \frac{\cos 2x - \cos 2\theta}{\cos x - \cos \theta} dx$

$I = \int \frac{(2\cos^2 x - 1) - (2\cos^2 \theta - 1)}{\cos x - \cos \theta} dx$ $\{\because \cos 2A = 2\cos^2 A - 1\}$

$I = \int \frac{2\cos^2 x - 2\cos^2 \theta}{\cos x - \cos \theta} dx$

$I = 2\int \frac{(\cos x - \cos \theta)(\cos x + \cos \theta)}{\cos x - \cos \theta} dx$

$I = 2\int (\cos x + \cos \theta) dx$

$I = 2[\sin x + x\cos \theta] + C$

39. (a)

Let $I = \int \sqrt{x^2 + 2x + 5}\, dx$

$I = \int \sqrt{(x+1)^2 + 4}\, dx$

$I = \frac{x+1}{2}\sqrt{(x+1)^2 + 4} + \frac{(2)^2}{2} \log |x + 1 + \sqrt{(x+1)^2 + 4}| + C$

$I = \frac{x+1}{2}\sqrt{x^2 + 2x + 5} + 2 \log |x + 1 + \sqrt{x^2 + 2x + 5}| + C$

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40. (b)

Let $I = \int_0^{\frac{\pi}{2}} \frac{\tan^7 x}{\cot^7 x + \tan^7 x} dx$ ... (1)

$I = \int_0^{\frac{\pi}{2}} \frac{\tan^7\left(\frac{\pi}{2} - x\right)}{\cot^7\left(\frac{\pi}{2} - x\right) + \tan^7\left(\frac{\pi}{2} - x\right)} dx$

$I = \int_0^{\frac{\pi}{2}} \frac{\cot^7 x}{\tan^7 x + \cot^7 x} dx$ ... (2)

Add equation (1) + (2)

$2I = \int_0^{\frac{\pi}{2}} \frac{\tan^7 x + \cot^7 x}{\tan^7 x + \cot^7 x} dx$

$2I = \int_0^{\frac{\pi}{2}} 1 \, dx$

$2I = [x]_0^{\pi/2}$

$I = \frac{\pi}{4}$

41. (a)

$I = \int_{-5}^{5} |x+2| dx$

$I = -\int_{-5}^{-2} (x+2) dx + \int_{-2}^{5} (x+2) dx$

$= -\left[\frac{x^2}{2} + 2x\right]_{-5}^{-2} + \left[\frac{x^2}{2} + 2x\right]_{-2}^{5}$

$= \left[-\left\{\frac{4}{2} - 4\right\} - \left\{\frac{25}{2} - 10\right\}\right] + \left[\left\{\frac{25}{2} + 10\right\} - \left\{\frac{4}{2} - 4\right\}\right]$

$= 29$

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42. (d)

$$I = \int_{-\pi/2}^{\pi/2} \frac{1}{e^{\sin x}+1}$$

Using property

$$\Rightarrow \int_{-a}^{a} f(x)dx = \int_{0}^{a} [f(x)+f(-x)]dx$$

$$\Rightarrow I = \int_{0}^{\pi/2} \left[\frac{1}{e^{\sin x}+1} + \frac{1}{e^{\sin(-x)}+1}\right]dx$$

$$I = \int_{0}^{\pi/2} \left[\frac{1}{e^{\sin x}+1} + \frac{e^{\sin x}}{e^{\sin x}+1}\right]dx$$

$$I = \int_{0}^{\pi/2} \frac{e^{\sin x}+1}{e^{\sin x}+1}dx$$

$$I = \int_{0}^{\pi/2} 1\,dx$$

$$I = [x]_0^{\pi/2}$$

$$I = \frac{\pi}{2}$$

43. (d)

$$I = \int_{0}^{\pi/2} \frac{1}{a^2\sin^2 x + b^2\cos^2 x}dx$$

$$I = \int_{0}^{\pi/2} \frac{\sec^2 x}{a^2\tan^2 x + b^2}dx$$

Let $\tan x = t$

$$\sec^2 x = \frac{dt}{dx}$$

$$\sec^2 x\,dx = dt$$

Now,

$$I = \int_{0}^{\infty} \frac{1}{a^2t^2 + b^2}dt$$

$$I = \frac{1}{a^2}\int_{0}^{\infty} \frac{1}{t^2 + \frac{b^2}{a^2}}dt$$

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$$I = \frac{1}{a^2} \times \frac{1}{b/a}\left[\tan^{-1}\frac{t}{b/a}\right]_0^{\infty}$$

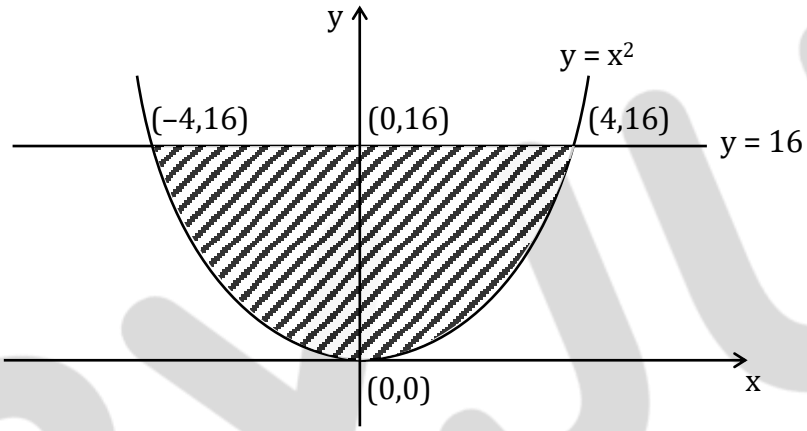
$$I = \frac{1}{ab}[\tan^{-1}\infty - \tan^{-1} 0]$$

$$I = \frac{1}{ab} \times \frac{\pi}{2}$$

$$I = \frac{\pi}{2ab}$$

44. (b)

Curve $y = x^2$ and line $y = 16$



$$\text{Area} = 2\int_0^{16} x\, dy$$

$$A = 2\int_0^{16} \sqrt{y}\, dy$$

$$A = 2\left[\frac{y^{3/2}}{3/2}\right]_0^{16}$$

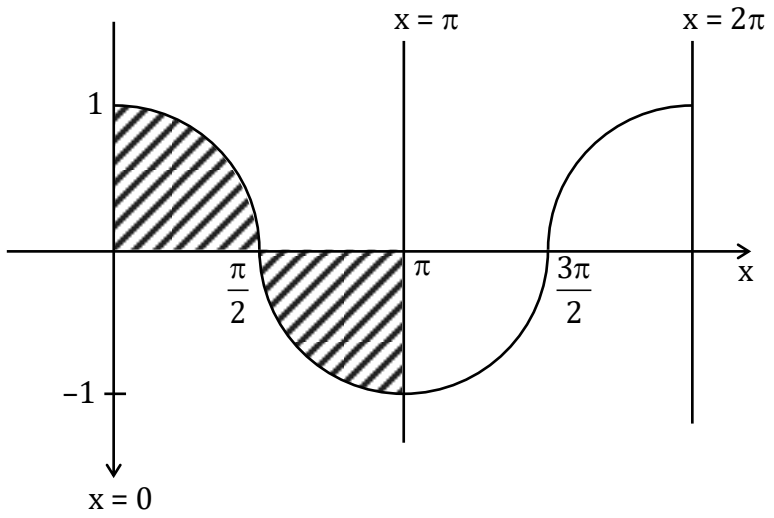
$$A = 2\left[\frac{(16)^{3/2}}{3/2}\right]$$

$$A = \frac{4}{3}(64)$$

$$A = \frac{256}{3} \text{ sq.units}$$

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45. (a)



y = cos x, x = 0 and x = π is shaded area

$$\text{Area} = \int_0^{\pi/2} \cos x \, dx + \int_{\pi/2}^{\pi} (-\cos x) \, dx$$

$$= \left[\sin x\right]_0^{\pi/2} - \left[\sin x\right]_{\pi/2}^{\pi}$$

$$= [1 - 0] - [0 - 1] = 2 \text{ sq. units}$$

46. (a)

$$\left[1 + \left(\frac{dy}{dx}\right)^2\right]^2 = \frac{d^2y}{dx^2}$$

The power of the term $\frac{d^2y}{dx^2}$ is 1.

Therefore, the degree of the given differential equation is 1.

47. (a)

$$\frac{dy}{dx} + y = 1$$

$$\Rightarrow \quad \frac{dy}{dx} = 1 - y$$

$$\Rightarrow \quad \frac{dy}{1-y} = dx$$

Integrate both side of the equation

$$\Rightarrow \quad \int \frac{dy}{1-y} = \int dx$$

$$\Rightarrow \quad -\log|1 - y| = x + C$$

$$\Rightarrow \quad \log\left|\frac{1}{1-y}\right| = x + C$$

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48. (a)

$$x\frac{dy}{dx} + 2y = x^2$$

$$\Rightarrow \quad \frac{dy}{dx} + \frac{2}{x}y = x$$

The integrating factor is calculated as

$$IF = e^{\int \frac{2}{x} dx}$$

$$= e^{2\log x}$$

$$= e^{\log x^2}$$

$$= x^2$$

49. (d)

Given vectors are

$$\vec{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$$

And $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$

Since, condition of orthogonal

$$\Rightarrow \quad \vec{a} \cdot \vec{b} = 0$$

$$\Rightarrow \quad (2\hat{i} + \lambda\hat{j} + \hat{k})\,(\hat{i} + 2\hat{j} + 3\hat{k})$$

$$\Rightarrow \quad 2 + 2\lambda + 3 = 0$$

$$2\lambda = -5$$

$$\lambda = -\frac{5}{2}$$

50. (c)

Because $\vec{a}, \vec{b}, \vec{c}$ are units vectors, therefore,

$$|\vec{a}| = 1, |\vec{b}| = 1 \text{ and } |\vec{c}| = 1 \qquad \ldots (1)$$

Again given $\vec{a} + \vec{b} + \vec{c} = \vec{0}$

Squaring both side

$$(\vec{a} + \vec{b} + \vec{c})^2 = 0$$

$$|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$$

$$1 + 1 + 1 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$$

$$2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = -3$$

$$\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{3}{2}$$

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51. (a)

Consider the unit vectors a, b and $(\sqrt{3}\,a - b)$

So, $(\sqrt{3}\,a - b) = 1$

Squaring both side

$\Rightarrow (\sqrt{3}\,a - b)^2 = 1$

$\Rightarrow 3|a|^2 + |b|^2 - 2\sqrt{3}\,a\cdot b = 1$

$\Rightarrow 3(1) + (1) - 2\sqrt{3}\,|a||b|\cos\theta = 1$

$\Rightarrow 4 - 2\sqrt{3}\cos\theta = 1$

$\Rightarrow 2\sqrt{3}\cos\theta = 3$

$\Rightarrow \cos\theta = \dfrac{\sqrt{3}}{2}$

$\Rightarrow \theta = \cos^{-1}\left(\dfrac{\sqrt{3}}{2}\right)$

$\Rightarrow \theta = \dfrac{\pi}{6} = 30°$

52. (d)

The reflection of any point (x, y, z) in xy-plane is (x, y, –z).

Therefore, the reflection of (α, β, γ) in xy-plane (α, β, –γ)

53. (c)

Plane equation: $2x - 3y + 6z - 11 = 0$

Plane: $\vec{n} = 2\hat{i} - 3\hat{j} + 6\hat{k}$

Line: $\vec{b} = \hat{i}$

Angle between line & plane = $\dfrac{\vec{n}\cdot\vec{b}}{|\vec{n}||\vec{b}|}$

$\Rightarrow \sin\theta = \dfrac{(2\hat{i} - 3\hat{j} + 6\hat{k})\cdot\hat{i}}{\left|\sqrt{(2)^2 + (-3)^2 + (6)^2}\right|\left|\sqrt{(1)^2}\right|}$

$\Rightarrow \sin\theta = \dfrac{2+0+0}{\sqrt{49}\sqrt{1}}$

$\Rightarrow \sin\theta = \dfrac{2}{7}$

$\Rightarrow \theta = \sin^{-1}\dfrac{2}{7}$

$\Rightarrow \sin^{-1}\alpha = \sin^{-1}\dfrac{2}{7}$

$\Rightarrow \alpha = \dfrac{2}{7}$

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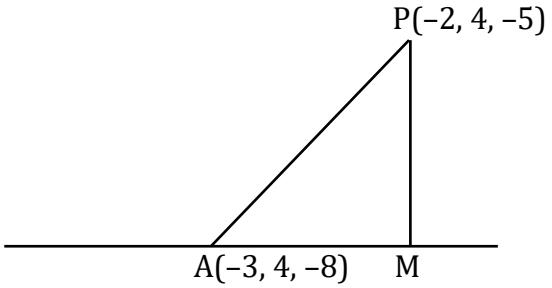
54. (b)

A line that passes through A (–3, 4, –8) and is parallel to the vector $b = 3\hat{i} + 5\hat{j} + 6\hat{k}$

Let M be the foot of the perpendicular from P (–2, 4, –5)

Then $AP = -\hat{i} - 3\hat{k}$

So $|AP| = \sqrt{1+9} = \sqrt{10}$



Therefore, AM is equal to the projection of AP on b

$$AM = \left|\frac{AP \cdot b}{|b|}\right|$$

$$= \left|\frac{(-\hat{i}-3\hat{k})\cdot(3\hat{i}+5\hat{j}+6\hat{k})}{|(3\hat{i}+5\hat{j}+6\hat{k})|}\right| = \left|\frac{-3-18}{\sqrt{9+25+36}}\right| = \frac{21}{\sqrt{70}}$$

Therefore

$$PM = \sqrt{AP^2 - AM^2}$$

$$= \sqrt{10 - \frac{441}{70}}$$

$$= \sqrt{\frac{259}{70}}$$

$$= \sqrt{\frac{37}{10}}$$

55. (c)

The probability of the defective ball is, $P = \frac{10}{100} = \frac{1}{10}$

The probability that the ball is not defective is,

$$Q = 1 - \frac{1}{10} = \frac{9}{10}$$

The probability the out of a sample of 5 pens drawn one by one without replacement and almost one is defective calculated as,

$$\text{Probability} = {}^5C_0\left(\frac{1}{10}\right)^0\left(\frac{9}{10}\right)^5 + {}^5C_1\left(\frac{1}{10}\right)^1\left(\frac{9}{10}\right)^4$$

$$= (1)\left(\frac{9}{10}\right)^5 + \left(\frac{5}{10}\right)\left(\frac{9}{10}\right)^4$$

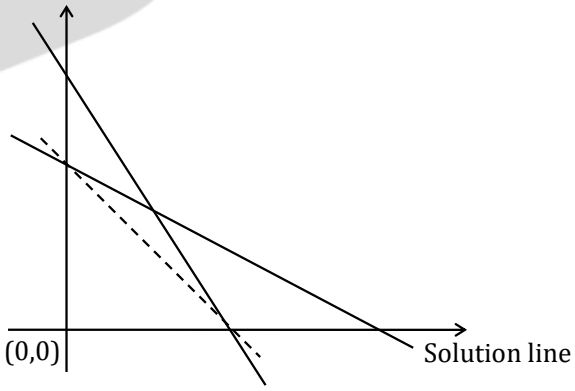
$$= \left(\frac{9}{10}\right)^5 + \frac{1}{2}\left(\frac{9}{10}\right)^4$$

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56. (b)
Two events A & B are said to be independent if,
$P(A' \cap B') = P(A') \cdot P(B')$
If A and B are independent then A' and B' are also independent.
Therefore,
$P(A' \cap B') = P(A') \cdot P(B')$
$= (1 - P(A))(1 - P(B))$
If A and B are mutually exclusive then, $P(A \cap B) = 0$
Check the options.

57. (a)
The sum of all probabilities is always equal to 1.
So,
$0.3 + K + 2K + 2K = 1$
$0.3 + 5K = 1$
$5K = 0.7$
$K = \frac{0.7}{5}$
$K = 0.14$

58. (c)
(1) Since, region (shaded) lies above x-axis.
$\therefore \quad y \geq 0$
(2) Shaded region is on left of $x = 6$
$\therefore \quad x \leq 6$
(3) Shaded region is below $y = 3$
$\therefore \quad y \leq 3$
(4) Equation of line passing through (4, 0) & (0, 5) is
$y - 0 = \frac{5}{-4}(x - 4)$
$5x + 4y - 20 = 0$
If $5x + 4y \geq 20$, put (0, 0) we get $0 \geq 20$
False
$\therefore$ Region lies away from origin which is true



$\therefore \quad 5x + 4y \geq 20$
(5) Shaded region lies on right side of y-axis
$\therefore \quad x \geq 0$
$\therefore \quad 5x + 4y \geq 20, x \leq 6, y \leq 3, x \geq 0, y \geq 0$

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59. (b)

When a LPP admits an optimal solution at two consecutive vertices of a feasible region.

Then the optimal solution occurs at every point on the line joining these two points.

60. (b)

Let $I = \int_{0.2}^{3.5} [x]\,dx$

$I = \int_{0.2}^{1} [x]\,dx + \int_{1}^{2} [x]\,dx + \int_{2}^{3} [x]\,dx + \int_{3}^{3.5} [x]\,dx$

$I = \int_{0.2}^{1} 0\,dx + \int_{1}^{2} 1\,dx + \int_{2}^{3} 2\,dx + \int_{3}^{3.5} 3\,dx$

$I = 0 + [x]_1^2 + 2[x]_2^3 + 3[x]_3^{3.5}$

$I = (2 - 1) + 2(3 - 2) + 3(3.5 - 3)$

$I = 1 + 2 + 1.5$

$I = 4.5$

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